

CryptoVault Pricing & Financial Analysis

This document outlines the business model, operational expenses, AWS hosting costs (in INR), and financial projections for **CryptoVault**.

1. Business Model Canvas (BMC)

Value Proposition

- **Risk-Free Simulated Trading:** High-fidelity paper trading engine mimicking real crypto exchange matching algorithms.
- **Advanced Order Book:** Supports real-time matching of Limit and Market orders.
- **Educational Value:** Safe, gamified environment with simulated funds (10,000 USDT) for beginner traders and groups.

Customer Segments

- **Beginner Crypto Traders:** Individuals who want to learn exchange operations without risking capital.
- **Trading Academies & Communities:** Educators seeking a clean sandbox environment to teach trading strategies.
- **Fintech Developers:** Developers experimenting with exchange APIs and WebSocket streams.

Revenue Streams

1. **Premium Subscription Plan:** ₹499/month per user for advanced features (custom API access, historical candles export, advanced indicator integrations).
2. **B2B White-Label Licensing:** ₹1,50,000 one-time setup fee + ₹25,000/month for custom-branded instances hosted for trading institutes.
3. **Ad Placements & Sponsorships:** ₹20,000/month for verified exchange recommendations and token promotions.

2. AWS Infrastructure Cost Breakdown (ap-south-1 Mumbai)

This projection represents a **Production-Ready tier** designed to support up to 10,000 concurrent active users.

Exchange rate used: 1 USD = ₹83.50 INR

AWS Service	Configuration	Monthly Cost (USD)	Monthly Cost (INR)
Application Server	2 × t3.medium EC2 Instances (Auto-Scaling)	\$41.76	₹3,487
Database Instance	Amazon DocumentDB db.t3.medium (with Replica)	\$57.60	₹4,810
Load Balancing	Application Load Balancer (ALB)	\$22.26	₹1,859
Storage & Transfer	100 GB EBS GP3 + 150 GB Data Transfer Out	\$24.00	₹2,004
Security & Backups	AWS Backup, CloudWatch, GuardDuty	\$10.00	₹835

Total Monthly Hosting	\$155.62	₹12,995
Total Annual Hosting	\$1,867.44	₹1,55,940

3. Financial Projections (Year 1)

These projections model target metrics assuming a user base of 1,000 registered accounts by Month 6, with a 5% premium conversion rate.

Monthly Operating Expenses (OPEX)

- **AWS Hosting Costs:** ₹12,995
- **Software Maintenance:** ₹35,000 *(Retainer for part-time developer support)*
- **Marketing & User Acquisition:** ₹15,000 *(Google/Meta ads targeting crypto educational groups)*
- **Administrative Overhead:** ₹3,000 *(Domain renewals, emails, auxiliary tools)*
- **Total Monthly Expenses:** ₹65,995
- **Total Annual Expenses:** ₹7,91,940

Projected Monthly Revenue (Targets)

- **Premium Subscriptions:** 50 subscribers × ₹499/month = ₹24,950
- **B2B Licensing Fees:** 2 active clients × ₹25,000/month = ₹50,000
- **Sponsorships & Ads:** ₹15,000
- **Total Monthly Revenue Target:** ₹89,950

Net Annual Summary (Year 1)

- **Gross Yearly Revenue:** ₹10,79,400 *(Includes upfront ₹1.5L setup fees for 2 B2B accounts)*
- **Total Yearly Expenses:** ₹7,91,940
- **Net Annual Profit:** ₹2,87,460
- **Profit Margin:** 26.6%